

напиши небольшой текст для рассказа вслух на английском

тема "Computer architecture and peripherals"

пиши не сложное, простые слова, который поймет b1

напиши 15 предложений

не пиши воды, приветственных фраз, риторических вопросов, просто текст наполненный информацией по теме

план твоего текста:

1. Введение в архитектуру компьютера
2. Классификация компьютеров
3. Основные компоненты компьютера
4. Развитие и тенденции в архитектуре компьютеров
5. Заключение

Tema N°1: «Computer architecture and peripherals».

Computer architecture is the way a computer system is built. It shows how the hardware and software work together. The main idea is to make computers fast, small, and efficient.

Computers can be divided into different types. Some are very powerful, like supercomputers. Others are smaller, like personal computers, laptops, and smartphones.

Every computer has some basic parts. The central processing unit, or CPU, is the brain of the computer. RAM stores temporary application data and instructions. Storage devices, like hard drives or SSDs, keep data for long-term use. Input devices, like a keyboard and mouse, help users give commands. Output devices, like a monitor or printer, show the results.

Over time, computer architecture has changed. CPUs now have many cores. Memory is faster and smaller. Computers use less power but do more work. New technologies like cloud computing and artificial intelligence also affect design.

In conclusion, computer architecture is always improving. It helps make better and faster computers for everyday use.

Tema N°2: «Networks».

A network is a group of computers connected together. They share data and resources like files, printers, and internet access. Networks help people work and communicate more easily.

There are two main types of networks: local area networks (LANs) and wide area networks (WANs). LANs connect computers in small areas, like schools or offices. WANs connect computers across cities or countries. The internet is the largest WAN.

Networks need special hardware. Routers and switches help move data between computers. A network card allows each computer to connect to the network. Cables or Wi-Fi are used to send signals.

Networks also need rules, called protocols. One important protocol is TCP/IP. It makes sure data goes to the right place. IP addresses help find each device on the network.

Networks are always growing. They are getting faster and more secure. In the future, networks will connect even more devices, like cars and smart homes.

Tema N°3: «Databases. Creating Databases for large organisations. Types and challenges».

A database is a system to store and manage data. Large organizations use databases to keep important information safe and organized. Databases help users find and update data quickly.

There are different types of databases. Relational databases store data in tables with rows and columns. NoSQL databases can store data in many formats like documents or graphs.

Creating a database for a large organization is complex. It needs to handle a lot of data and many users at the same time. The database must be fast and reliable. It also needs strong security to protect sensitive information.

One challenge is keeping the data accurate. Another problem is backing up data to prevent loss. Large databases also require good design to avoid slow performance. Managing databases needs skilled professionals and special software.

In conclusion, databases are important for big organizations. But creating and managing them can be difficult because of size, speed, and security needs.

Tema N°4: «Types of Operating systems. Advantages and disadvantages».

Operating systems are software that manage computer hardware and programs. They help users interact with the computer easily.

There are several types of operating systems for different devices. One type is desktop operating systems, like Windows, macOS, and Linux. They are good for personal computers

and offer many programs. Another type is mobile operating systems, like Android and iOS. These are made for smartphones and tablets.

There are also server operating systems, used in large computers that manage networks and websites. Real-time operating systems work in machines that need fast responses, like robots or cars.

Each type has advantages and disadvantages. Desktop systems have many features but need good hardware. Mobile systems are easy to use and portable but have less power. Server systems are very reliable but can be expensive. Real-time systems are fast but hard to design.

In conclusion, different operating systems fit different needs. Choosing the right one depends on the device and tasks.

Tema N°5: « BMSTU/ My department and my specialty (future professional career)/My Major».

I study at Bauman Moscow State Technical University. My department is IU5, which focuses on computer science. My major is Informatics and Computer Engineering.

This major includes both software and hardware. We study programming, algorithms, databases, and computer systems. It helps us understand how computers work and how to create new technologies.

I already work as a backend developer. I write server-side code using the Go programming language. My job is to create the backend part of web services.

I also write firmware for MultiSIM CPE devices. These devices connect to the internet using different mobile networks. The firmware controls how these devices work.

My work combines programming, networking, and system design. I use the knowledge from my university in real projects. In the future, I want to grow as a backend engineer and work on large-scale systems.

Tema N°6: «Software and Operating systems».

Software is a set of instructions that tells a computer what to do. It includes programs, applications, and operating systems. Software helps users perform tasks on a computer.

There are different types of software. System software includes the operating system and tools that help run the computer. Application software includes programs like word processors, games, and web browsers.

An operating system is a special software. It controls the computer's hardware and manages other software. Without an operating system, the computer cannot work properly.

Operating systems manage files, memory, and devices like printers and keyboards. They also provide a user interface, so people can interact with the computer easily.

In conclusion, software and operating systems are important parts of computing. The operating system manages hardware and software, while other software helps users complete tasks.